



Tecnológico de Monterrey
Interpartial Exam 2 A
2026-Jan-May

Grade

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Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 09/02/2026

Last Names and Names: _____

Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Limits. Properties and Algebraically.

1. (6 points) Compute the limit of the function $f(x) = \frac{x^2-4}{x-2}$ as $x \rightarrow 2$.

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2. (7 points) Compute the limit of the function $f(x) = \frac{6x^2+4}{5x^2-2}$ as $x \rightarrow \infty$.

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3. (7 points) Compute the limit of the function $f(x) = \frac{\sqrt{x+4}-5}{x}$ as $x \rightarrow 0$.

II Derivative using the power rule and properties of derivatives.

1. (10 points) Using the power rule, compute the derivative of the function $f(x) = \frac{3}{2x}$.

2. (10 points) Using the power rule, compute the derivative of the function $g(x) = 4 \left[\left(\frac{x}{2} \right)^{12} - \left(\frac{x}{2} \right)^6 \right]$.

III Derivatives of trascendental functions.

- 1. (6 points)** Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = \sin(x) + \cos(x)$.

- 2. (6 points)** Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = e^x + \ln(x)$.

- 3. (8 points)** Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = \ln(5x)$.

IV Derivative by definition. $\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

1. (20 **points**) Compute the derivative of the function $f(x) = \sqrt{x}$ using the definition of derivatives.

2. (20 **points**) Compute the derivative of the function $f(x) = |x|$ at $x = 0$ using the definition of derivatives.